Dexter Carpenter

Mechatronics Engineer

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Overview

Mechatronics Engineer with a strong foundation in mechanical engineering and over a year of professional experience in industrial automation. I've developed a multidisciplinary skillset spanning mechanical design, electrical systems, and controls engineering, with a focus on rapidly delivering functional, scalable solutions in fast-moving environments. Through both industry experience and academic projects, I've taken on complex technical challenges and contributed to full system development from concept through implementation. I'm driven to continue growing as an engineer while delivering reliable and scalable solutions in the automation space.

Experience

Mechatronics Engineer, Timberline Automation & Control Co. (TAC) Dec. 2024 – Apr. 2026

- Worked amongst a team of 4 to execute full-stack automation solutions from conception to commission
- Designed and built 8 integrated mechanical, electrical, and controls systems

Control Panels & PLC Software for Autonomous Package Unloading System

- Owned design of UL-listed electrical control panels and PLC architecture with 120 I/O points for a robotic truck unloading system, enabling successful deployment in a live Fortune 500 distribution facility
- Engineered recovery logic to detect and handle dropped or missed packages during robotic picking, maintaining continuous operation and minimizing manual intervention during runtime
- Enabled dynamic picking logic to operate without a predefined truck manifest, allowing the system to identify and process unknown packages in real time and achieve >1200 cases per hour throughput

Project Lead, OPEnS Lab *FloDar* Jan. 2021 – Nov. 2022

- Worked to continue the development of *FloDar*, an environmental sensor to monitor groundwater discharge and its properties.

Projects

Rocket Air Brakes for Oregon State Competition Rocket Team Sept. 2023 – Jun. 2024

- Co-developed a rocket air brake system, including mechanical design, control algorithms, and simulation tools to reduce the rocket's apogee for a better competition score.

Honda Civic CR-X Electric Vehicle Conversion Dec. 2025 – Present

- Actively working with peers to convert a 1987 Honda Civic to an EV using Nissan Leaf donor vehicle

Public Obsidian Notes Repository Jul. 2025 – Present

- Developed a public-facing repository that selectively syncs content from a private Obsidian vault using GitHub Actions

Education

Bachelor of Science (B.S.), Mechanical Engineering Sept. 2019 – Aug. 2024

Oregon State University, Corvallis, OR | 2024

Skills

- Control System Design
- PLC/HMI Programming
- Hardware-Agnostic Simulation
- Computer Networking
- Git
- Control Panel Design

Applications

- Microsoft Office
- Google Workspace
- Monday
- Slack
- TwinCAT
- GitHub

Programming Languages

- IEC 61131-3 (Structured Text)
- C#/C++
- Python
- MATLAB
- HTML/CSS
- Ruby

I encourage you to check out my website!